In most ACL-based systems, these components are not explicitly separated. The resource often acts as both PAP and PDP—storing its own access list and enforcing decisions based on it. The PIP is minimal, usually just resolving the current user's identity or group membership. While simple and efficient, this tight coupling can make ACLs difficult to scale across distributed systems or centralize for governance.

5.1.3 Strengths and limitations of ACLs

Access Control Lists have stood the test of time for a reason: they're simple, direct, and easy to understand. Because ACLs attach permissions directly to the resource, it's usually clear who can access what. That clarity can be valuable in small environments or systems with a strong ownership model.

However, this simplicity comes at a cost. As the number of users and resources grows, managing permissions individually becomes unmanageable. Table 5.2 summarizes the most common strengths and limitations of ACLs. This comparison highlights why ACLs remain appealing in some contexts but are problematic in others, especially as systems scale and require more adaptive access decisions.

Table 5.2 Strengths and limitations of Access Control Lists (ACLs)

Strengths	Limitations
Easy to understand and implement	Becomes difficult to manage in large, complex environments
Permissions are local and transparent	No centralized or global view of access
Fine-grained control on individual resources	Hard to apply consistent policies across many resources
Fast to evaluate at runtime	High administrative overhead in dynamic environments
Familiar and widely supported in legacy systems	Not well-suited to modern, scalable access control
Predictable and deterministic access decisions	Inflexible—doesn't support contextual or conditional access

In short, ACLs work well when access control needs are relatively static, local, and clear, especially in systems that support inheritance or hierarchical scoping to reduce repetitive configuration. However, as systems evolve—supporting multiple users, teams, tenants, or compliance constraints—